

Semisolid dosage forms

Oct, 2022

Chapter Learning Objectives.

At the end of this chapter students will be able to

- Describe the physical properties, method of preparation, packaging and labeling, storage and uses of ointments, pastes, creams and gels
- Describe the formulation excipients used
- Discuss the advantages and disadvantages of ointments, pastes, creams and gels.
- Define transdermal drug delivery system
- Discuss factors affecting percutaneous drug absorption, its advantage and disadvantage.

Introduction

Definition:

- ✓ Semi solids are the topical dosage forms used for the therapeutic, protective or cosmetic function.
- ✓ Pharmaceutical semisolid dosage preparations include :
 - ❖ ointments,
 - ❖ pastes,
 - ❖ cream,
 - ❖ Gels etc.

Introduction.....

- Semisolid dosage forms usually are intended for **localized drug delivery.**
 - They can be applied topically to the skin, cornea, rectal tissue, nasal mucosa, vagina, buccal tissue, external ear lining ,etc
- They also have been explored for the systemic delivery of various drugs.

Introduction.....

Ideal Properties Of Semisolid Dosage Forms

- **Physical Properties**

Smooth texture

Elegant in appearance

Non dehydrating

Non gritty

Non greasy and non staining

Non hygroscopic

Introduction.....

Physiological Properties

- Non irritating
- Do not alter membrane / skin functioning
- Miscible with skin secretion
- Have low sensitization index

Ingredients used in semisolid preparation:

- They contain one or more active ingredients dissolved or uniformly dispersed in a suitable base and any suitable excipients such as:
 - Bases
 - Preservative
 - Humectants
 - Antioxidants
 - Emulsifier
 - Gelling agent
 - Permeation enhancer
 - Buffers

Bases

- It is one of the most important ingredient used in formulation of semisolid dosage form.
- Ointment bases **do not merely act as the carriers of the medicaments**, but they **also control the extent of absorption of medicaments** incorporated in them.

Ideal properties of a base

They should be:

- Inert, non-irritating and non-sensitizing.
- Compatible with skin pH and the drug.
- Good solvent and/or emulsifying agent.
- Emollient, protective, non-greasy and easily removable.
- Release medicament readily at the site of application.
- Pharmaceutically elegant and possess good stability



OINTMENTS & PASTES

Pharmaceutical ointments and paste

- Pharmaceutical ointments are semisolid systems that are applied externally
 - ✓ primarily to the skin
 - ✓ mucous membranes e.g. the rectum, the vagina, the eye.
- **Medicated ointments:** for the t/t of infection, inflammation .
- **Non-medicated ointments** are commonly used as emollient/
lubricating properties.

Pharmaceutical ointments and paste

- Pharmaceutical pastes are generally composed of ointment bases that contain a high concentration (frequently 50% w/w) of dispersed drug.
- The viscosity of pharmaceutical pastes is greater than that of pharmaceutical ointments.

Advantage of semi-solid dosage form

- It is used **externally**
- Probability of side effect **can be reduced**
- **First pass gut and hepatic metabolism is avoided.**
- **Local action and Site specific action of drug on affected area.**
- Convenient for **unconscious patient or patient having difficulty on oral administration.**
- Suitable dosage form for **bitter drugs.**
- **More stable than liquid dosage form**

Advantages of pharmaceutical ointments, pastes and gels

- Ointments easily spread on skin, being retained at the site of application as an occlusive layer.
- There by preventing moisture loss from the skin.
- useful in restoration of the physical characteristics of the skin (e.g. due to inflammation).
- Ointments are associated with lubricating properties that may be employed to reduce trauma of an affected site upon spreading.

Advantages.....

- The increased viscosity of pharmaceutical pastes ensures that a thick film of the dosage form is applied to the site of action.
 - shows excellent persistence.
- Due to the high solids content, Paste
 - act to absorb moisture and chemicals within the exudates.
 - enables to be used as a sun block.

Advantages.....

- The chemical stability of therapeutic agents that are prone to hydrolysis will be dramatically enhanced by formulation within pharmaceutical ointments and pastes.
- Pharmaceutical gels may be formulated to provide excellent spreading properties and will provide a cooling effect due to solvent evaporation.

Disadvantages of semi-solid dosage form

- There is **no dosage accuracy** in this type of dosage form
- The base which is used in the semi-solid dosage form can be **easily oxidized**.
- May cause **staining**.
- They are **bulky to handle**.
- Application with finger may cause **contamination**.
- Physico-chemically **less stable than solid dosage** form.
- May cause **irritation or allergy** to some patients

Disadvantages of ointments, pastes and gels

- Ointments are generally greasy and difficult to remove (cosmetically unacceptable).
- Pharmaceutical pastes are generally applied as a thick layer at the required site (cosmetically unacceptable).
- Staining of clothes is the problem
- Problematic in ensuring spreading of the dosage form over the affected site.

Disadvantages...

- Pharmaceutical ointments may not be applied to exuding sites
- Problems concerning drug release from pharmaceutical ointments may occur
 - if the drug has limited solubility in the ointment base
- Pharmaceutical pastes are generally not applied to the hair
 - due to difficulties associated with removal.
- Drugs that are prone to hydrolysis should not be formulated into aqueous gels.

Ointments

Classification of Ointments

- ✓ **Based on penetration**
- ✓ **According to their therapeutic uses.**

Ointments.....

i. Based on penetration

1. Epidermic ointments

- **Act on epidermis & produce local effect.**
- **Used as protectives, antiseptic, local anti-infectives & parasiticides.**

2. Endodermic ointments

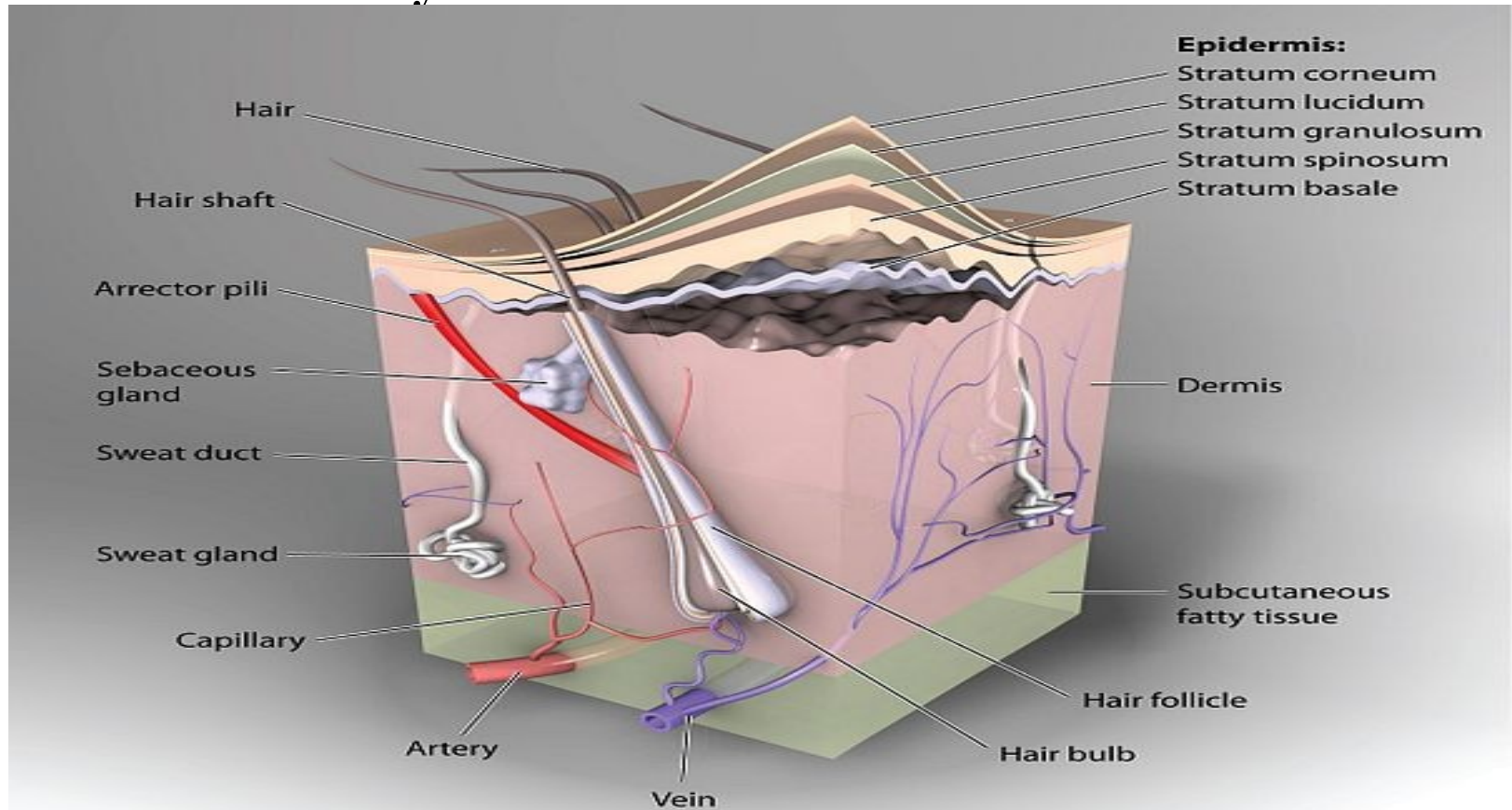
- **Act on deeper layers of cutaneous tissues.**
- **Partially absorbed & act as a emollients, stimulants local irritants.**

3. Diadermic ointments

- **Meant for deep penetration and release the medicaments and produce systemic effects.**

Skin structure.....

- Skin consists of three layers Epidermis, dermis and subcutaneous layer



Ointments.....

ii) According to their therapeutic uses. :

1. Antibiotic ointments:

- Used to kill micro-organisms.

Eg- bacitracin, Neomycin, Chlortetracyclines, etc.

2. Antifungal ointments- Inhibit or kill the fungi.

Eg- Benzoic acid, salicylic acid, nystatin etc.

3. Anti-inflammatory ointments: Relieve inflammatory, allergic.

Eg- Betamethasone valerate, Hydrocortisone & its acetates.

Ointments.....

4. Anti-pruritic ointments: Relieve itching

Eg-Benzocaine & coal tar

5. Astringent ointments:

-Causes contraction of skin & decreases discharge.

Ex- Calamine, ZnO, Acetic acid, Tannic acid

6. Anti eczematous Ointments:

-Prevent oozing & excretion from vesicles on the skin.

Ex- Hydrocortisones, ichthamol, coal tar & salicylic acid

Ointments.....

7. Keratolytic Ointments:

- Used to remove or soften the horny layer of the skin.

Eg- Resorcinol, salicylic acid & sulphur

8. Counter-irritant Ointments:

- Applied locally to irritate skin, thus reducing or relieving another irritation or deep seated pain.

Eg- methyl salicylate, iodine, oleoresin

9. Protectant Ointments:

- Protect skin from moisture, air, sun rays, chemicals.

Ex- Calamine, ZnO, silicones, titanium dioxide etc.

Ointments.....

10. Antidandruff ointments:

Eg- Salicylic acid, cetrimide.

11. Ointment For Psoriasis treatment:

Eg- coal tar, corticosteroid, & salicylic acid mixed with suitable ointment base.

12. Parasiticide ointments:

- Destroy or inhibit living infestation like ticks & lice.

Eg- Benzyl benzoate, hexachloride, sulphur etc.

Ointments.....

- The formulation of ointments and pastes involves the **dispersal or dissolution of the selected therapeutic agent into an ointment base.**
- **The physicochemical properties of the ointment base are fundamental to:**
 - **the clinical**
 - **non-clinical performance of this type of dosage form.**

Ointments.....

The choice of ointment base is dependent on several factors

1. Dermatological factors

- **Absorption and penetration**
- **Effect on the skin**
- **Miscibility with skin secretion**
- **Compatibility with skin secretion**
- **Non-irritant**
- **Emollient property**
- **Patient skin condition (old, baby and young)**
- **the site of application;**

Ointments.....

2. Pharmaceutical factors

- the required rate of drug release;
- the chemical stability of the drug; and
- the effect of the therapeutic agent on formulation viscosity.

Types of base for ointments and pastes

- There are four types of base that are used to formulate pharmaceutical ointments and pastes:

(1) hydrocarbon;

(2) absorption;

(3) water-miscible/removable; and

(4) water-soluble

Types of base.....

1. Hydrocarbon bases

- Oleaginous bases
- non-aqueous formulations, emollient
- restrict water loss from the site of application
 - **Due to the formation of an occlusive film**
- Excellent retention on the skin
- Predominantly hydrophobic
 - Difficult to remove from the skin by washing
 - difficult to apply to (spread over) wet surfaces (e.g. mucous membranes, wet skin)

Hydrocarbon bases.....

- Small amount of water (5%) can be incorporated into it with difficulty
- Can be protective to water labile drugs such as tetracycline and bacitracin.
- Greasy and can stain clothing.
- Chemically inert

Hydrocarbon bases....

Examples of hydrocarbon bases:

1. **Petrolatum, USP**

- Yellow petrolatum/petrolatum jelly /Vaseline
- Melts at 38-60°C

2. **White petrolatum, USP**

- Decolored petrolatum,
- White petroleum jelly/white vaseline

Hydrocarbon bases....

3. Yellow ointment, USP

Yellow beeswax (5% w/w) + petrolatum (95% w/w)

4. White ointment, USP

White beeswax + white petrolatum

Hydrocarbon bases frequently contain the following components:

- (1) **hard paraffin;**
- (2) **white/yellow soft paraffin; and**
- (3) **liquid paraffin (mineral oil)**

Hydrocarbon bases.....

Hard paraffin

- A mixture of solid saturated hydrocarbons that are derived from petroleum or shale oil.
- a colourless or white wax-like material that is physically composed of a mixture of microcrystal.
- The melting temperature of hard paraffin is 47 -65°C
- used to enhance the rheological properties of ointment bases.

Hydrocarbon bases.....

White/yellow soft paraffin

- A purified mixture of semisolid hydrocarbons that are derived from petroleum.
- Consists of microcrystals embedded in a gel composed of liquid and amorphous hydrocarbons.
- The melting range of the soft paraffins is 38-60°C.

Hydrocarbon bases.....

- White soft paraffin and yellow soft paraffin (the former being a bleached form of yellow soft paraffin)
 - used as an ointment base without the need for additional components, although it may be combined with liquid paraffin

Hydrocarbon bases....

Liquid paraffin (mineral oil, Liquid petrolatum)

- mixture of refined liquid saturated hydrocarbons obtained from petroleum.
- Levigating agent to incorporate lipophilic solids
- An excipient in topical formulations where its emollient properties are exploited
- Formulations containing liquid paraffin require the incorporation of an antioxidant
 - due to the ability of this material to **undergo oxidation**.

2. Absorption bases

- Absorption bases are hydrophilic and, therefore, can absorb considerable amounts of water or aqueous solutions.

I. Non-emulsified bases

- These bases absorb water and aqueous solutions to produce water-in- oil (**W/O**) emulsions.

✦ Compared with the hydrocarbon bases:-

- They are **less occlusive**, nevertheless, are **good emollients**
- They assist oil-insoluble medicaments to penetrate the skin
- They are easier to **spread**.

Absorption base....

- Typically non-emulsified bases are commonly composed of:
 - one/more paraffins, and
 - a sterol-based emulsifying agent.
- Examples: lanolin (wool fat); wool alcohols; and beeswax (white or yellow).

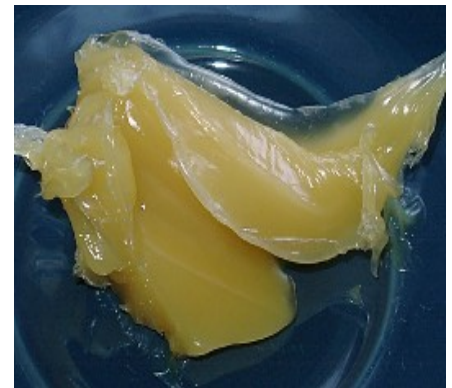
Absorption bases...

Lanolin (wool fat)

- Lanolin is a wax-like material that is derived from sheep's wool.
- It is available in two forms, termed **Anhydrous lanolin (wool fat) and hydrous lanolin.**

Anhydrous lanolin:

- called **wool wax, wool fat, or wool grease,**
- a greasy yellow substance
- contains $< 0.25\%$ of water



Absorption bases.....

- Typically mixed with paraffins to produce an ointment base
 - Can absorb approximately twice its own weight of water to produce water in oil emulsions.
 - **The usual concentrations of lanolin used in ointments (e.g. Simple Ointment BP) range from 5 to 10% w/w.**

Absorption bases.....

- Wool fat is a major constituent of **Simple Ointment B.P** and **Eye Ointment B.P**

Simple ointment B.P

- Hard paraffin50g
- Cetosteryl alcohol....50g
- **Wool fat**.....50g
- Soft paraffin.....850g

Absorption bases.....

Eye Ointment B.P

Liquid paraffin.....100g

Wool fat.....100g

Yellow soft paraffin.....850g

Absorption bases.....

Wool alcohols

- A crude mixture of sterols and triterpene alcohols
- Added to mixtures of paraffins (hard, or white/yellow soft or liquid) to produce the required consistency.
- **The inclusion of wool alcohols (5% w/w) results in a 300% increase in the concentration of water**
eg. wool alcohol ointment B.P

Absorption bases.....

Beeswax (white or yellow)

- A wax that consists of esters of aliphatic alcohols (C_{24} – C_{36}) and linear aliphatic fatty acids (up to C_{36}) that is combined with paraffins to **produce non-emulsified bases**.
- **White beeswax is the bleached form of yellow beeswax.**
- Included in some ointment bases to increase water-absorbing power.

Absorption bases.....

II. Water in oil emulsions

- These are similar in properties to the previous group and are **capable of absorbing water**.
- Absorption bases are **less occlusive** than the hydrocarbon bases and easier to **spread**.
- They are **good emollients**.
 - Eg. **hydrous lanolin**, which is a **mixture of lanolin and 25–30% water**.

Absorption bases.....

- It is used alone as an **emollient** and is an ingredient of several B.P.C ointment bases, i,e
 - Hydrous wool fat ointment
 - Calamine and coal tar ointment
 - Methylsalicylate ointment

Absorption bases.....

Properties of absorption bases

- **Non-emulsified absorption bases (anhydrous)**
 - Emollient
 - Occlusive
 - Absorbs water
 - Greasy
- **W/O emulsion**
 - Emollient
 - Occlusive
 - Contains water, absorbs additional water
 - Greasy

Water-washable/removable bases

- Water-miscible bases that are used to form **O/W** emulsions.
- They can be applied to **scalp** and other **hairy regions**.
- There are 3 official **anhydrous water-miscible bases**.
 - Emulsifying ointment B.P – anionic
 - Cetrimide emulsifying ointment B.P – cationic
 - Cetomacrogol emulsifying B.P- non-ionic

Water-washable/removable bases.....

○ Advantages of water-miscible bases

- ✓ able to accommodate large volumes of water**
- ✓ able to accommodate excess moisture**
e.g. exudate from abrasions & wounds.
- ✓ Reduced interference with skin function**
- ✓ easily washed from the skin and from clothing.**
- ✓ readily applied to (and removed from) hair, skin.**

Hence High cosmetic acceptability

Water-washable/removable bases....

N.B. Ointments with **hydrocarbon** or **absorption** bases are not very suitable for **scalp** conditions because their removal is unpleasant and difficult.

Water removable bases.....

Characteristics of water removable bases

- **Resemble creams in their appearance**
- **May be diluted with water or with aqueous solution**
- **Certain medicinal agents may be better absorbed in the skin**
- **Water washable**
- **Contains water**
- **Can absorb water**
- **Non-occlusive**
- **Non-greasy**

Water-soluble bases

- Completely water-soluble bases have been developed from the Macrogols (Polyethylene glycols), a range of compounds with the general formula:-



Water-soluble bases.....

- Polyethylene glycols are polymers of **ethylene oxide** and **water**

Polycondensation



- The chain length may be varied to achieve polymers having desired **viscosity** and **physical form**
 - Macrogols 200,300 ,400 - **Viscous liquid**
 - Macrogol 1500 - **Semi-solid**
 - Macrogols 6000 - **waxy solids**

Water-soluble bases.....

Example:

- **PEG ointment, NF**
 - **PEG 3350400 g**
 - PEG 400 600 g**

Water-soluble bases.....

Advantages of water- soluble bases

- **Water solubility**
- Freedom from greasiness
 - Easily removal from the skin
 - Readily miscible with tissue exudates
- **Good absorption by the skin**
 - valuable when drugs are required to penetrate the skin
- **Good solvent properties**
 - For some water soluble dermatological e.g.

Water-soluble bases.....

Disadvantages of water-soluble bases

- Less bland than paraffins, possibly due to their **hygroscopic nature.**
- Reduction in activity of certain antimicrobial agents, **due to hydrolysis**

water-soluble bases, disadvantage.....

- Inability to incorporate large volumes of aqueous solutions b/c
 - Ointments will soften or
 - dissolve the ointment base if the concentration of water is large enough (5% w/w).
- Its use is usually reserved for the incorporation of solid therapeutic agents.
- It may incorporate up to **25% of an aqueous solution** if a portion of the lower-molecular-weight polyethylene glycol is replaced with stearyl alcohol.

Properties of water-soluble bases

- **Water soluble and washable**
- **Non-greasy**
- **Non/less occlusive**
- **Lipid free**
- **Synthetic base**
- **Relatively inert**

Summary of ointment bases

Properties	Oleaginous bases	Absorption bases	Water-washable bases	Water soluble bases
Solubility in water	Insoluble in water	Insoluble in water	Insoluble in water	Soluble in water
Washability	Not water washable	Not water washable	water washable	Water washable
Absorption of water	Not absorb water	absorbs water	absorbs water	Absorbs water To the pt of solubility
Examples	Hard paraffin, soft paraffin, liq. paraffin	Wool fat, wool alcohol, lanolin, bees wax	Cold creams, Emollient creams Hydrophilic ointment	Macrogols Liq. PEG 200

Miscellaneous excipients

Miscellaneous excipients of ointments and pastes

- **the therapeutic agent may be added as a solid component**
- **In absorption and water-miscible bases, the addition may be in the form of a solution.**
- **This solution may be aqueous, alcoholic (e.g. propylene glycol, glycerol) or hydroalcoholic**
 - ❖ **must not adversely affect the physical stability and/or the appearance**

Miscellaneous excipients.....

- Other excipients may be included in ointments and pastes, including:
 - (1) additional/alternative solvents;**
 - (2) preservatives; and**
 - (3) antioxidants.**

Miscellaneous excipients.....

Additional/alternative solvents

- Are hydrophobic liquid components that may be added to ointment bases (predominantly hydrophobic or absorption bases).
- **Examples of these include:**
 - (1) liquid silicone;**
 - (2) vegetable oils;**
 - (3) Polar organic solvents, and**
 - (4) organic esters**

Miscellaneous excipients.....

Liquid silicone (polydimethylsiloxane)

- used in barrier ointments (water-repellent properties) to protect the skin against water-soluble irritants

Vegetable oils

- used to replace mineral oils
- added to hydrophobic or absorption bases to increase the emollient properties.

Miscellaneous excipients.....

Disadvantages:

Prone to **oxidation** requiring the inclusion of antioxidants, stabilizers which may cause sensitization in some patients.

E.g. coconut oil, arachis oil,

Polar organic solvents

- used to disperse small quantities of active medicament though a relatively large quantity of ointment base.

Miscellaneous excipients.....

Organic esters

- used to replace a mineral oil to enhance the spreadability
- to enhance drug dissolution within the ointment base.
- more consistent in composition than the vegetable oils and are more **resistant to oxidation.**

Eg. isopropyl myristate (most commonly used)

Miscellaneous excipients.....

❖ Antioxidants

- The antioxidants that are used in ointment bases are similar to those listed for emulsion
 - Butylated hydroxy anisole (BHA)
 - Butylated hydroxytoluene (BHT)
 - ethyl, propyl or dodecyl gallates
- Chelating agents such as ethylene diamine tetra-acetic acid (EDTA) may also be used.

Miscellaneous excipients.....

■ Preservatives

- Preservative may not be required in anhydrous ointments because **the substrate** is generally unfavorable to the multiplication of any **contaminating micro-organisms**.
- Ointment with an aqueous component requires effective **antimicrobial agents** to prevent the growth of organisms that may cause **spoilage and pathogenicity**.

Miscellaneous excipients.....

- The preservative for emulsions are also suitable for **hydrous ointment bases**.
- Those used most commonly are mixtures of hydroxybenzoate esters, sorbic acid, phenethyl alcohol, organic mercurials and quaternary ammonium compounds.
- The effectiveness of the preservative system should be established by **challenge tests** with appropriate

Method of preparation

- A well-made ointment is:-
 - a). **Uniform throughout**, i.e. it contains no lumps of separated high melting pt ingredient of the base.
 - b). **Free from grittiness**, insoluble powders are finely subdivided and large clumps of particles are absent.
- Methods of preparation must try to satisfy these

Method of preparation.....

- Both on a **large** and a **small scale**, ointments are prepared by two general methods.
 - **Incorporation**
 - **Fusion**
- The method for particular preparation depends primarily up on the **nature of ingredients.**

Method of preparation.....

I. Incorporation method

- ❑ In the incorporation method **the component of the ointment are mixed together by various means until a uniform preparation has been attained.**
- ❑ **Before incorporation the ingredient should have the finest state.**

Method of preparation, Incorporation method...

❑ On a **small scale**, as in the extemporaneous compounding of Rx:

➤ **Mortal and pestle**

➤ **Ointment tile (Slab) and spatula** : are a large glass or porcelain plate may be used to rub the ingredients together.

Method of preparation

Incorporation method...

- **Mortar and Pestle**
 - **Used:**
 1. when large quantities of liquids are to be incorporated.
 2. when large quantity of ointment is to be prepared



Method of preparation, Incorporation method...

- Otherwise, **tile** and **spatula** is satisfactory due to its large surface area.
- On a **large scale** manufactured in stainless steel tank.



Method of preparation, Incorporation method...

1. Incorporation of solids

- When preparing an ointment by **spatulation**, the pharmacist generally works the ointment with a stainless steel spatula with a long, broad blade and periodically removes the accumulation of ointment on the **larger spatula** with a **smaller spatula**.
- If the component of an ointment are reactive with the metal of the spatula (eg, phenol), hard **rubber spatula** may be used.

Method of preparation, Incorporation of solids...

- ❑ The ointment is prepared by thoroughly **rubbing** and **working** the component together on the hard surface with the spatula until the product is smooth and uniform.
- ❑ Generally the **ointment base** is placed on **one side** of the working surface, and the powdered components, previously reduced to fine powders and thoroughly blended in the mortar, are placed on the other.

Method of preparation,
Incorporation of solids...

- ❑ Then a portion of powder is mixed with a portion of the base until uniform, and the process is repeated until all portions of the powder and base are combined.
- ❑ The portions of prepared ointment are then combined and thoroughly blended by continuous & movement of the spatula and through the combined portions of ointment.

Method of preparation, Incorporation of solids...

- When only a small portion of powder is to be added, it may be added in its entirety to a small portion of ointment base.
- If the quantity of active ingredient is very small you have to use levigating agent (mineral oil, and glycerin)
- After levigation, the dispersion is incorporated with the remainder of the base by spatulation or by using the mortar and pestle

Method of preparation...

2. Incorporation of liquids

- ❑ Any liquids ingredients should be incorporated at the end of levigation.
- ❑ Generally, **mortar and pestle** is preferred when large volumes of liquid are added than an ointment slab.

Method of preparation...

Fusion method of preparation

- All or some of the components of an ointment are combined by being **melted** together and **cooled** with **constant stirring** until congealed
- When ointment base contains several ingredients of **different melting point**; first add constituent of high melting point and then adding in **descending order of melting point**.

Method of preparation...

Fusion method...

- Advantages of melting in descending order of melting point:
 - ▶ Quicker
 - ▶ Require less heat
 - ▶ Avoid over heating or easily melted constituent.

Method of preparation...

Fusion method...

- Those components **not melted** are generally added to the congealing mixture as its is being cooled and stirred.
- Naturally, **heat-labile substances** and **any volatile** components are added last when the temperature of the mixture is low enough not to cause **decomposition** and **volatilization** of the component.

Method of preparation...

Fusion method...



- ✚ Many substances are added to the congealing mixture in **solution**, others are added as **insoluble powders** generally levigated with a portion of the base.
- ✚ On a **small scale**; the fusion process may be conducted in a porcelain dish or glass beaker
- ✚ On a **large scale**; it is generally carried out in large steam-jacketed kettles.

Method of preparation...

Fusion method...

- ✚ Once congealed, the ointment may be passed through:
 - ✓ an ointment mill (in large-scale manufacture)
 - ✓ rubbed with a spatula or in a mortar (in small scale ppn) to ensure a **uniform texture**.
- ✚ Many medicated ointments and ointment bases containing such components as beeswax, hard paraffin, stearyl alcohol, and high molecular weight polyethylene glycols, which do not tend themselves well to mixture by incorporation, are prepared by **fusion**.

Method of preparation...

Fusion method...

- ✚ After melting; the melted ointment base should be stirred taking care, not to cause localized cooling of high melting ointment constituents .
- ✚ Localized cooling can be occurred due to:-
 - Cold spatula or stirrer
 - Putting the dish on cold surface
 - Transferring melted base to cold dish

Method of preparation...

Fusion method...

- ✚ If the product is **granular** after cooling, it has to be remelted with minimum amount of heat and stirred gently until it sets fully.
- ✚ We use fusion method when:-
 - **Constituents are with hard consistency (high melting point components).**
 - **The medicaments is soluble in melted base.**

Difficulties encountered in preparation of ointment

- ❖ Danger of **chemical reaction** between steel spatula and some reactive ingredients.

Ex. I_2 , salicylic acid, Benzoic acid.....

Solution :- to minimize this problem, use **bone spatula, wooden spatula or hard rubber spatula**.

- ❖ Incorporation of aqueous solution in to hydrocarbon base.

Solution: addition of wool fat.

- ❖ Incorporation of antibiotics sensitive to hydrolysis

Solution:- use hydrocarbon base.

Packaging and storage of ointment

❖ Ointments are usually packaged either:-

➤ **Jars**

➤ **Tubes**

Jars : - May be made of glass, uncolored, dark green, amber, or blue

- Or made of plastic, or porcelain.



Packaging and storage of ointment...

- ❖ Tubes:- are made of tin or of plastic (collapsible tube)



Packaging and storage of ointment...

- ❖ Ointment jars may be filled on a small scale by packaging the weighed amount of ointment in to the jar by means of a **flexible spatula and forcing the ointment** down and along the sides of the jar to avoid the entrapment of air.
 - ❖ Ointments prepared **by fusion** may be poured directly in to the ointment jars for congealing with in the jar.
- These oints. normally assume a finished look

Packaging and storage of ointment...

- ❖ In the **large-scale** manufacture of ointments, **pressure fillers** force a specified amount of an ointment in to a jar.
- ❖ Tubes are generally filled **by pressure fillers** from the open back end (opposite and from the cap end) of the tube which is then closed and sealed.

Packaging and storage of ointment...

- ❖ Tube-filled ointment predominate over jar-filled ointments primarily b/s they are more convenient for the patient.
- Ointments in tubes are less exposed to air and to potential contaminants and are therefore likely to be more stable
 - ✓ remain efficacious for longer periods of time
- ❖ Most oints must be stored at temperature below 30°C to prevent the softening and liquefying.

2. PASTES

- Pastes, like ointments, are intended for external application to the skin.
- They differ from ointments primarily in that they containing **large proportion of solid materials**
 - as a consequence are **thicker** and **stiffer** than ointment.
- Pastes are used principally as
 - absorbents,
 - antiseptics
 - protective, or to soothe broken skin surfaces,

Pastes...

- They are **emollient** but, because of the powder content, **porous**, hence, perspiration can escape.
- The powder **absorb exudates**, less macerating than ointments with similar base.
- Are **less greasy** than ointments because of the **absorption of the fluid hydrocarbon fraction** to the insoluble particles.

Pastes...

- Most are **unsuitable for treating scalp** conditions because they are **difficult** to remove from the hair.
- Because of the **stiffness** and **absorptive qualities** of pastes, **they remain in place after application** with little tendency to soften and flow
 - therefore effectively employed to absorb secretion from the site of application.

Pastes...

- Pastes are therefore preferred for acute lesions that have a tendency toward **crusting, or oozing**.
- The film formed on application is opaque and thus can often serve as a **sun-block**.
- **Pastes are prepared similarly to ointments.**
 - However, when a **levigating agents** is to be used to render the powdered component smooth, **a portion of the base** is often used rather than a liquid like mineral oil

Pastes...

- Among the few pastes in use today is zinc oxide paste (Lassar's Plain Zinc Paste),
 - prepared by levigating and then mixing 25% each of zinc oxide and starch and calamine with white petrolatum.
- The product is very firm and is better able to protect the skin and absorb secretions than is zinc oxide ointment

Zinc oxide paste, USP

- | | |
|--------------------|-----------|
| ○ Zinc oxide | 25.0% |
| ○ Starch | 25.0% |
| ○ Calamine | 5.0% |
| ○ White petrolatum | q.s. 100% |

Procedure:

- Titrate the calamine with the zinc oxide and starch and incorporate uniformly in the petrolatum by levigation in a mortar or on a glass slab with a spatula.
- **Mineral oil should not be used as a levigating agent,**
- A portion of petrolatum can be melted and used as a levigating agent if so desired.

- The term ‘cream’ in pharmacy and medicine is applied to **viscous emulsion or semi-solid emulsion preparation** consisting of solutions or dispersions of one or more medicaments in suitable base and intended for application to the skin or mucous membrane.
- The term cream is widely used in **pharmaceutical** and **cosmetic industry**.

Creams...

- ❖ Many patients and physicians prefer creams to ointments.
 - they are generally easier to spread, and, in the case of cream of the oil-in-water emulsion.
 - Easier to remove than many ointments
- ❖ They are applied to the skin for:-
 - Protective
 - Beautifying
 - Therapeutic or prophylactic purp



Creams...

- ❖ Creams may contain suitable antimicrobial or **preservatives** unless the medicaments or basis have sufficient intrinsic bactericidal and fungicidal activity
- ❖ Creams are mainly two types:-
 - 1). Water- in –oil (W/O) - oily cream
 - 2). Oil –in-water (O/W) - aqueous cream
- ❖ **O/W emulsions are most useful as water-washable bases, whereas w/o emulsions are emollient and cleansing.**

Creams...

- ❖ Patients often prefer O/W b/s
 - Cream **spreads** more readily.
 - Is less **greasy**
 - Evaporating water **soothes** the inflamed tissue.

Creams...

- ❖ **O/W (vanishing cream)** After application of the cream the water evaporates leaving behind a thin residue film of the stearic acid.
 - For **oozing** or **weeping surface** (wound that release exudates from skin) = b/s O/W mix oozing surface.
- ❖ **Vanishing cream** is prepared by incorporation of significant amount of stearic acid in an oil-in-water emulsion.

Creams...

- ❖ Sodium or potassium hydroxide reacts with **the stearic acid** to form a soap emulsifier.
- ❖ **Humectants** (glycerin, PEG, 7% sorbitol) are frequently added to vanishing creams and **O/W emulsions** to prevent drying out.

General compounding procedure for creams

- Emulsified creams are prepared by heating the components of the oily phase (usually including the emulgent) until molten and then cooling to 60⁰c .
- **The component of the aqueous phase are mixed in a separate vessel and also heated to 60⁰c.**
- The aqueous phase is then added to the oily phase at the same temperature.

General compounding of creams...

- This is important and a thermometer should be used.
 - ✓ The resulting emulsion should be stirred until cool .
- Rapid cooling may result in separation of high melting point components.
- Excessive aeration caused by vigorous stirring may also lead to a granular product.

General compounding of creams...

- If necessary the product may be homogenized after cooling.
- Creams may contain one or more medicaments in solution in one or other of the phases.
- Finely powdered insoluble medicaments may be dispensed in a cream base

Container of Creams



- **Wide-mouthed squat jars** may be used for creams where the risk of contamination in use is considered to be minimal, e.g. Oils creams.
- The container must be well closed and prevent water evaporation.
- The mouth of the jar should be covered with a disc of **grease proof paper**.

Container of Creams...

- **Collapsible metal or flexible plastic tubes** are to be preferred since these reduce the risk of contamination in use and most proprietary products are packed in tubes.



Gels (gelies)



- Gels are transparent or translucent semi-solid or solid preparations, consisting of solutions or dispersions of one or more active ingredients in suitable hydrophilic or hydrophobic bases.
- Semisolid preparations that contain small inorganic particles or large organic molecules interpenetrated by a liquid.

Gels (gelies)...

- Gels made of inorganic materials are usually **two - phase systems** where small discrete particles are dispersed throughout the dispersion medium.
- When the particle size of the dispersed phase is larger, they are referred to as **magmas**.
- Gels made of organic molecules **are single - phase systems**, where no apparent physical boundary is seen between the dispersed phase and the dispersion medium.

Gels (gelies)...

The dispersion medium:

- aqueous
- Hydroalcoholic or oleaginous base
- Gels are attractive delivery systems:
 - ❖ simple to manufacture
 - ❖ suitable for administering drugs through skin, oral, buccal, ophthalmic, nasal, otic, and vaginal routes
 - ❖ provide intimate contact between the drug and the site of action or absorption.

Gels (gelies)...

- Gels exhibit different physical properties, namely, imbibition, swelling, syneresis, and thixotropy
- **Thixotropy:** *non* -Newtonian flow nature of gels, which is characterized by a reversible gel - to - sol formation with no change in volume or temperature
- **Imbibition-** is the taking up of a certain amount of liquid by a gel without a measurable increase in volume.

Gels (gelies)...

- **Swelling**-is the taking up of a liquid by a gel with an **increase** in volume.
- **Syneresis**-Syneresis refers to the contraction or shrinkage of gels as a result of squeezing out of dispersion medium from the gel matrix.
- Syneresis is a form of **instability** in aqueous and non aqueous gels

Classification and Types of Gels

- based on the physical properties of the gelling agent in the dispersion.
- hydrogels
- organogels

1. Hydrogels: prepared with water - soluble materials or water - dispersible colloids.

Eg. Natural and synthetic gums such as tragacanth, sodium alginate, and pectin inorganic materials: alumina, bentonite, silica, and veegum organic materials such as cellulose polymers

Classification and Types of Gels

2.Organogels (oleaginous gels): prepared using water - insoluble oleaginous materials.

- Prepared using **water - insoluble lipids** such as glycerol esters of fatty acids, **which swell in water**
Eg. Glycerol monooleate, glycerol monopalmitate, and glycerol monolinoleate

Components in gel formulation

Gelling agent

Gelling agent causes thickening

➤ are either organic hydrocolloids or hydrophilic inorganic substance.

- **They are water dispersible, possess swelling properties, and improve the viscosity of dispersions.**

Components in gel formulation ...

An ideal gelling agent

- Not interact with other formulation components
- Be free from microbial attack.
- Changes in the temperature and pH during pps and preservation should not alter its rheological properties.
- Economical
- Readily available
- Colorless gels,
- provide cooling sensation on the site of application, and possess a pleasant odor.

Examples of Gelling Agents

1. Acacia
2. Bentonite
3. Carboxymethylcellulose Na
4. Colloidal silicon dioxide
5. Gelatin
6. Hydroxyethylcellulose
7. Hydroxypropylmethylcellulose
8. Maltodextrin
9. Polyvinyl alcohol
10. Propylene carbonate
11. Alginic acid
12. Carbomer
13. Cetostearyl Alcohol
14. Ethylcellulose
15. Guar gum
16. Hydroxypropyl cellulose
17. Magnesium Al silicate
18. Methylcellulose
19. Povidone
20. Sodium alginate
21. Sodium starch glycolate
22. Starch
23. Tragacanth
24. Xanthan gum

Components in gel formulation ...

- **Preservatives**
 - Methylhydroxybenzoate
 - Propylhydroxybenzoate
- **Hygroscopic agents (Humectant)**
 - **Eg.** Glycerol, propylene glycol or sorbitol solution.
- **Chelating agents**
 - For ingredients which are sensitive to heavy metals. eg.
EDTA.

Components in gel formulation

Flavours/sweetening agents

- Included in pharmaceutical gels that are designed for administration into the **oral cavity for the treatment of infection, inflammation or ulceration.**

Buffers

- As in other pharmaceutical formulations, buffers (e.g. phosphate, citrate) may be included in aqueous& hydroalcoholic-based gels

Colorants

Magmas

- Are aqueous suspensions of insoluble, inorganic drugs and **differ from gels mainly in that the suspended particles are larger.**
- **When prepared, they are thick and viscous, so need of a suspending agent.**

Preparation of Gels and Magmas

- **Generally, the water soluble components are initially dissolved in the vehicle in a mixing vessel with stirring.**
- **The hydrophilic polymer must be added to the stirred mixture slowly to prevent aggregation then continuing stirring until dissolution of the polymer**
 - **excessive stirring of pharmaceutical gels results in entrapment of air.**
 - **Therefore, to prevent this the mixing rate must not be excessive.**

Preparation of Magmas and Gels

- ❖ By freshly precipitating the disperse phase (chemical rxn)
 - ✓ The desired gelatinous precipitate results when solutions of **inorganic agents react to form an insoluble chemical having a high attraction for water.**
- **Eg:** Milk of magnesia is a preparation containing 7 to 8.5% magnesium hydroxide.

Preparation of Magmas and Gels...

- It may be prepared by a reaction between sodium hydroxide and magnesium sulfate.



- ❖ By direct hydration of inorganic chemical in water.

Example: Hydration of Magnesium oxide



Gels (gelies)....

- Use: for **medication**, **lubrication** and some **miscellaneous** application
- **Medicated Jellies**
 - Jellies contain a considerable amount of water and are particularly suitable as vehicle for water-soluble medicaments such as local anesthetics, spermicides, and antiseptics.
- **Lubricants Jellies**
 - Lubricants for - glove
- **Miscellaneous jellies**
 - Patch testing : As a vehicle for allergens applied to the skin to detect sensitivity

Examples of Magmas and Gels

Bentonite Magma	NF	suspending agent
Sodium Fluoride and Phosphoric Acid Gel	USP	dental care prophylactic
Fluocinonide Gel	USP	Anti-inflammatory corticosteroid
Tretinoin Gel	USP	treatment for acne
Erythromycin and Benzoyl peroxide Gel		
Clindamycin Topical Gel		
Hydroquinone Gel		Hyperpigmented skin
Salicylic acid Gel		keratolytic
Desoximethasone Gel		anti-inflammatory and antipruritic
Aluminum Phosphate Gel (Amphogel)	USP	antacid
Aluminum hydroxide Gel	USP	antacid
Dihydroxyaluminum Aminoacetate Magma	USP	antacid
Milk of Magnesia (Magnesia Magma)	USP	Antacid; laxative

Containers

- ❖ Containers should be well-filled, to minimize evaporation of water into the air space, **well closed** and **stored in a cool place, to prevent drying out.**

▪

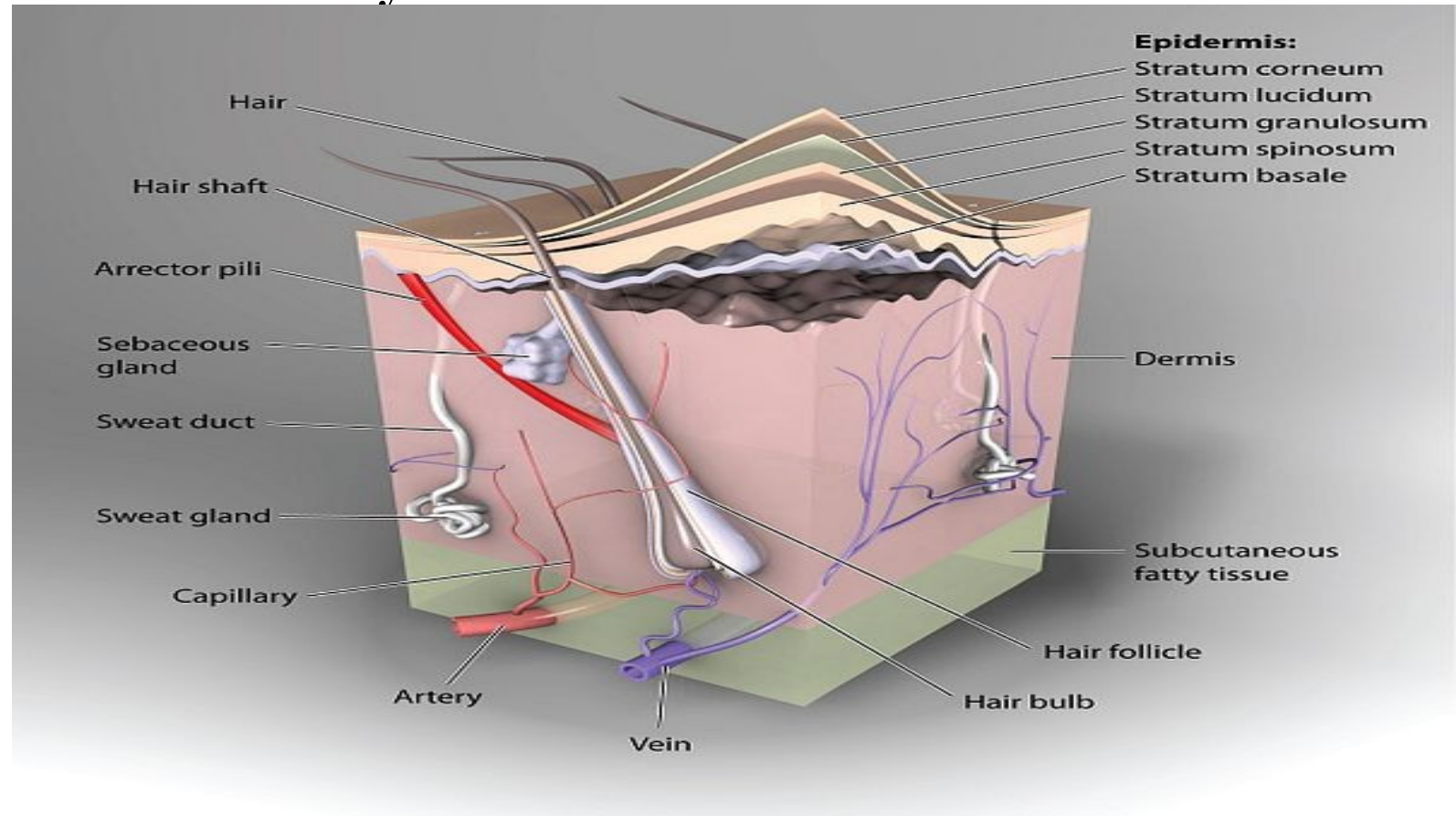
Percutaneous absorption (Transdermal drug delivery(TDD))

Percutaneous absorption (Transdermal drug delivery)

- ❑ **Percutaneous absorption** : the absorption of substances from outside the skin to positions beneath the skin, including entrance into the blood stream.

Skin structure.....

- Skin consists of three layers Epidermis, dermis and subcutaneous layer

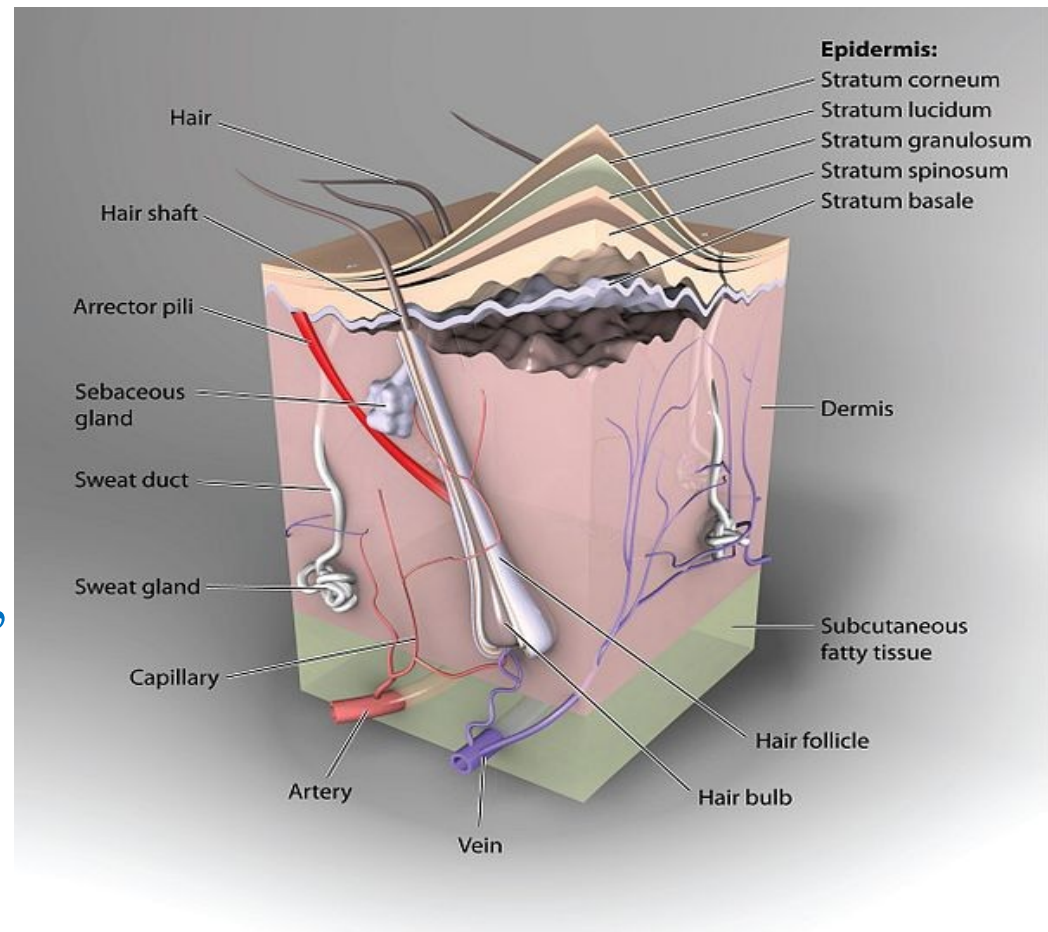


Percutaneous absorption...

○ Drugs may penetrate intact skin after topical

application through

- the walls of the hair follicles
- the sweat glands
- the sebaceous glands,
- Trough the cells of the horny layer



Percutaneous absorption.....

- **The main barrier for the penetration of drug is generally through the epidermal layers.**
- The stratum corneum is the outermost ‘horny’ layer of skin, comprising about partially desiccated, keratinized epidermal cells.
- **It is the rate-limiting barrier to percutaneous drug transport**

Percutaneous absorption.....

Drug transport in the skin involves

- Dissolution and release of the drug from the formulation
- drug partitioning into the sc
- drug diffusion across sc
- drug partitioning from sc into viable epidermis layer
- Diffusion across the viable epidermis layer into the dermis
- Drug absorption by capillary which achieves systemic circulation

Percutaneous absorption.....

- Transport **of lipophilic drug molecules** is facilitated
 - by their dissolution into **intercellular lipids**
- Absorption of **hydrophilic molecules** through skin can occur through ‘**pores**’ or **openings of the hair follicles** and **sebaceous glands**.
 - **But, the relative surface area of these openings is barely 1% of the total skin surface.**
 - **limits the amount of drug absorption**

Percutaneous absorption.....

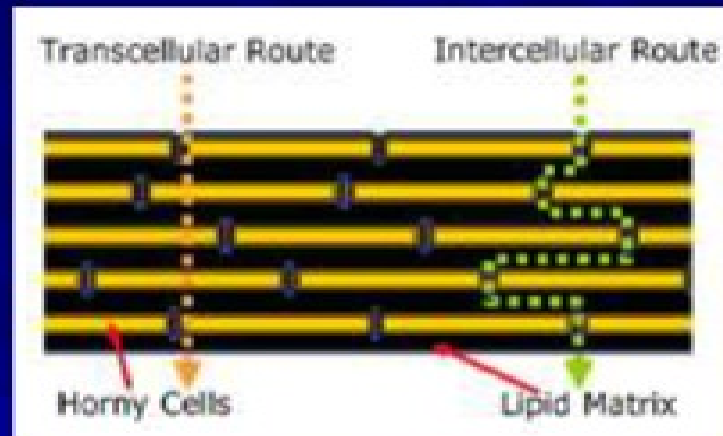
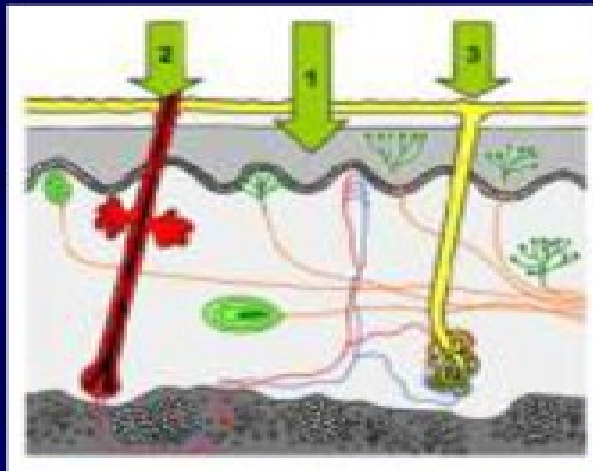
- **Permeation of the substances in SC can occur by diffusion via:**

1. Transcellular penetration

2. Intercellular penetration

Percutaneous absorption.....

Pathways of drug penetration through skin



•Simplified diagrams showing routes of drug penetration.

(a) **Macroroutes of drug penetration** (1) across the continuous stratum corneum; (2) through the hair follicles with their associated sebaceous glands or (3) via the sweat duct.

(b) Representation of the stratum corneum membrane, illustrating two possible **microroutes for permeation**. (1) Intercellular (2) transcellular

Factors influencing percutaneous absorption

I. Skin conditions

1. Damage and disease

- Intact skin presents a barrier to absorption that can be reduced considerably when the skin is damaged or is in a diseased state
- Skin can be damaged by dryness, irritation, allergic reactions or by abrasion
- Certain acids and alkalis may injure barrier cells

Factors influencing, skin conditions.....

- In heavy industry, workers' skins may lose their reactivity or 'harden' because of frequent contact with irritant chemicals.
- **Many solvents open up the complex dense structure of the horny layer**

Factors influencing, skin conditions.....

2. Age

- **The skin of the young and the elderly is more permeable than adult.**
- Children are more susceptible to the toxic effects of drugs and chemicals.
 - partly because of their greater surface area per unit body weight.

Factors influencing, skin conditions.....

Potent topical steroids, boric acid and hexachlorophane have produced severe side-effects and death.

- **Premature infants may be born with no stratum corneum.**
- **Advantageous to treat breathing difficulties with caffeine or pain with buprenorphine, via simple topical application.**

Factors influencing, Skin conditions.....

3. Blood flow

- Changes in the peripheral circulation could affect transdermal absorption.
- An increased blood flow could raise the conc. gradient across the skin.
 - Increase percutaneous absorption

Factors influencing, Skin conditions.....

4. Skin metabolism

- The skin metabolizes steroid hormones, chemical carcinogens and some drugs.
- Such metabolism may determine the therapeutic efficacy of topically applied compounds (particularly prodrugs).
- It has been estimated that the skin can metabolize some 5% of candidate topical drugs.

Factors influencing, Skin conditions.....

5. Skin hydration

- When water saturates the skin the tissue swells, softens and wrinkles permeability increases markedly.
- The keratinized cells of the stratum corneum have the ability to hold water.
- **Hydration may result from water diffusing from underlying epidermal layers from perspiration that accumulates after the application of an occlusive vehicle or dressing.**

Eg. use of occlusive plastic films in topical steroid treatment
(**penetration increases ten fold**).

Factors influencing, Skin conditions.....

6. Temperature

- Absorption is influenced by environmental factors such as skin temperature and surface humidity.
- Diffusion can be accelerated by raising surface temperature.
- **Occlusive vehicles increase skin temperature** by a few degrees, but any consequent increased permeability is small compared to the effect of hydration.

II. Physicochemical characteristics of the drug

1. The permeability constant, P

- ✓ Is the product of stratum corneum / vehicle **partition coefficient, K_m** and **degree of diffusion, D** of the compound in the skin. h is thickness of the membrane

$$\frac{K_m D}{h} = P \quad \text{Permeability constant}$$

- ✓ A substances that are very soluble in a particular vehicle will tend to have a low permeability constant and will therefore be poorly absorbed.
- ✓ **The lipid-soluble unionized species of many substance are more rapidly absorbed than their water- soluble salts.**

Factors influencing, Physicochemical characteristics.

2) Molecular modification

- Molecular modification of active substances can have marked effects on their activity.
- **Changes in functional groups** that alter the solubility and partition coefficient of substance b/n the vehicle and skin barrier.

Factors influencing, Physicochemical characteristics .

3) Concentration

- An increase in the concentration of an active substance increase in the quantity of the material absorbed.

4) Molecular size

- Molecules of small size in high concentration tend to penetrate more readily than large molecules

Factors influencing, Physicochemical characteristics....

5) Particle size

- Particle size reduction has been used to enhance the percutaneous absorption of **poorly soluble drugs** such as hydrocortisone or hydrocortisone acetate.

Factors influencing percutaneous absorption

III. Miscellaneous Factors

- Site of application
- Time of contact
- Amount of preparation employed
- pH of applied preparation

Maximizing the bioavailability of drug to skin:

- Drug or prodrug selection
- Hydration
- Ultrasounds(phonophoresis)
- Iontophoresis
- Stratum corneum removal
- Chemical penetration enhancers

Chemical penetration enhancers

Materials used to enhance absorption:

- surfactants, ozone, dimethylsulfoxide (DMSO), dimethylacetamide, dimethylformamide, alcohol, acetone, propylene glycol, and polyethylene glycol.

Mechanism of action for percutaneous absorption enhancers

- Reduction of the resistance of the stratum corneum
- Alteration of the hydration of the stratum corneum
- Effecting a change in the structure of the lipids and lipoproteins in the cellular channels, through denaturation
- Carrier mechanism in the transport of ionizable drugs

Iontophoresis and Sonophoresis.....

- **Iontophoresis:** the delivery of charged chemical compounds across the skin membrane using an applied electrical field.

Eg: lidocaine, amino acids/peptides/insulin, verapamil, and propranolol

- **Sonophoresis** is also being studied as a means to enhance transdermal drug delivery

Eg.: hydrocortisone, lidocaine, and salicylic acid in such formulations as gels, creams and lotions

Advantage of TDD

- Avoids GI drug absorption difficulties
- Substitutes for oral administration of medication when that route is unsuitable
- Avoids first-pass effect
- Provides sustained and controlled administration
- Provides capacity to terminate drug effect rapidly
- Permit self administration

Disadvantage of TDD

- unsuitable for drugs that irritate or sensitize the skin.
- Only relatively potent drugs are suitable candidates
- Technical difficulties are associated with the adhesion of the systems to different skin types and under various environment conditions
- Poor diffusion of large molecules

?????